Practical No 04. Advanced Layout and View Handling

A) Develop an Android application using ScrollView to display long scrollable content.

Project Name: ScrollViewDemo

Program Code:

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:fillViewport="true"
  android:clipToPadding="false"
  tools:context=".MainActivity">


  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:paddingStart="16dp"
    android:paddingEnd="16dp"
    android:paddingTop="10dp"
    android:paddingBottom="20dp">

    <TextView
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:gravity="center"
      android:paddingBottom="10dp"
      android:text="STUDENT REGISTRATION FORM"
      android:textColor="#4A4A4A"
      android:textSize="20sp"
      android:textStyle="bold" />

    <EditText
      android:id="@+id/editTextFullName"
      android:layout_width="match_parent"
      android:layout_height="45dp"
```

```
    android:hint="Enter Full Name"
    android:inputType="textPersonName" />

  <EditText
    android:id="@+id/editTextEmail"
    android:layout_width="match_parent"
    android:layout_height="45dp"
    android:hint="Enter Email"
    android:inputType="textEmailAddress" />

  <EditText
    android:id="@+id/editTextMobile"
    android:layout_width="match_parent"
    android:layout_height="45dp"
    android:hint="Enter Mobile Number"
    android:inputType="phone" />

  <EditText
    android:id="@+id/editTextAddress"
    android:layout_width="match_parent"
    android:layout_height="45dp"
    android:hint="Enter Address"
    android:inputType="textPostalAddress" />

  <EditText
    android:id="@+id/editTextCity"
    android:layout_width="match_parent"
    android:layout_height="45dp"
    android:hint="Enter City"
    android:inputType="text" />

  <EditText
    android:id="@+id/editTextState"
    android:layout_width="match_parent"
    android:layout_height="45dp"
    android:hint="Enter State"
    android:inputType="text" />

  <EditText
    android:id="@+id/editTextCountry"
    android:layout_width="match_parent"
    android:layout_height="45dp"
    android:hint="Enter Country"
    android:inputType="text" />

  <EditText
    android:id="@+id/editTextPincode"
```

```
        android:layout_width="match_parent"
        android:layout_height="45dp"
        android:hint="Enter Pincode"
        android:inputType="number" />

    <EditText
        android:id="@+id/editTextQualification"
        android:layout_width="match_parent"
        android:layout_height="45dp"
        android:hint="Enter Qualification"
        android:inputType="text" />

    <EditText
        android:id="@+id/editTextCollege"
        android:layout_width="match_parent"
        android:layout_height="45dp"
        android:hint="Enter College Name"
        android:inputType="text" />

    <EditText
        android:id="@+id/editTextCourse"
        android:layout_width="match_parent"
        android:layout_height="45dp"
        android:hint="Enter Course"
        android:inputType="text" />

    <EditText
        android:id="@+id/editTextSkills"
        android:layout_width="match_parent"
        android:layout_height="45dp"
        android:hint="Enter Skills"
        android:inputType="text" />

    <EditText
        android:id="@+id/editTextExperience"
        android:layout_width="match_parent"
        android:layout_height="45dp"
        android:hint="Enter Experience"
        android:inputType="text" />

    <EditText
        android:id="@+id/editTextHobby"
        android:layout_width="match_parent"
        android:layout_height="45dp"
        android:hint="Enter Hobby"
        android:inputType="text" />
```

```
    <EditText
      android:id="@+id/editTextAge"
      android:layout_width="match_parent"
      android:layout_height="45dp"
      android:hint="Enter Age"
      android:inputType="number" />

    <EditText
      android:id="@+id/editTextDOB"
      android:layout_width="match_parent"
      android:layout_height="45dp"
      android:hint="Enter D.O.B."
      android:inputType="text" />

    <EditText
      android:id="@+id/editTextGender"
      android:layout_width="match_parent"
      android:layout_height="45dp"
      android:hint="Enter Gender"
      android:inputType="text" />

    <EditText
      android:id="@+id/editTextFather"
      android:layout_width="match_parent"
      android:layout_height="45dp"
      android:hint="Enter Father's Name"
      android:inputType="textPersonName" />

    <EditText
      android:id="@+id/editTextMother"
      android:layout_width="match_parent"
      android:layout_height="45dp"
      android:hint="Enter Mother's Name"
      android:inputType="textPersonName" />

    <EditText
      android:id="@+id/editTextParentMobile"
      android:layout_width="match_parent"
      android:layout_height="45dp"
      android:hint="Enter Parent's Mobile Number"
      android:inputType="phone" />

    <EditText
      android:id="@+id/editTextNationality"
      android:layout_width="match_parent"
      android:layout_height="45dp"
      android:hint="Enter Nationality"
```

```
    android:inputType="text" />

<EditText
    android:id="@+id/editTextBloodGroup"
    android:layout_width="match_parent"
    android:layout_height="45dp"
    android:hint="Enter Blood Group"
    android:inputType="text" />

<EditText
    android:id="@+id/editTextUsername"
    android:layout_width="match_parent"
    android:layout_height="45dp"
    android:hint="Enter Username"
    android:inputType="text" />

<EditText
    android:id="@+id/editTextPassword"
    android:layout_width="match_parent"
    android:layout_height="45dp"
    android:hint="Enter Password"
    android:inputType="textPassword" />

<EditText
    android:id="@+id/editTextConfirmPassword"
    android:layout_width="match_parent"
    android:layout_height="45dp"
    android:hint="Confirm Password"
    android:inputType="textPassword" />

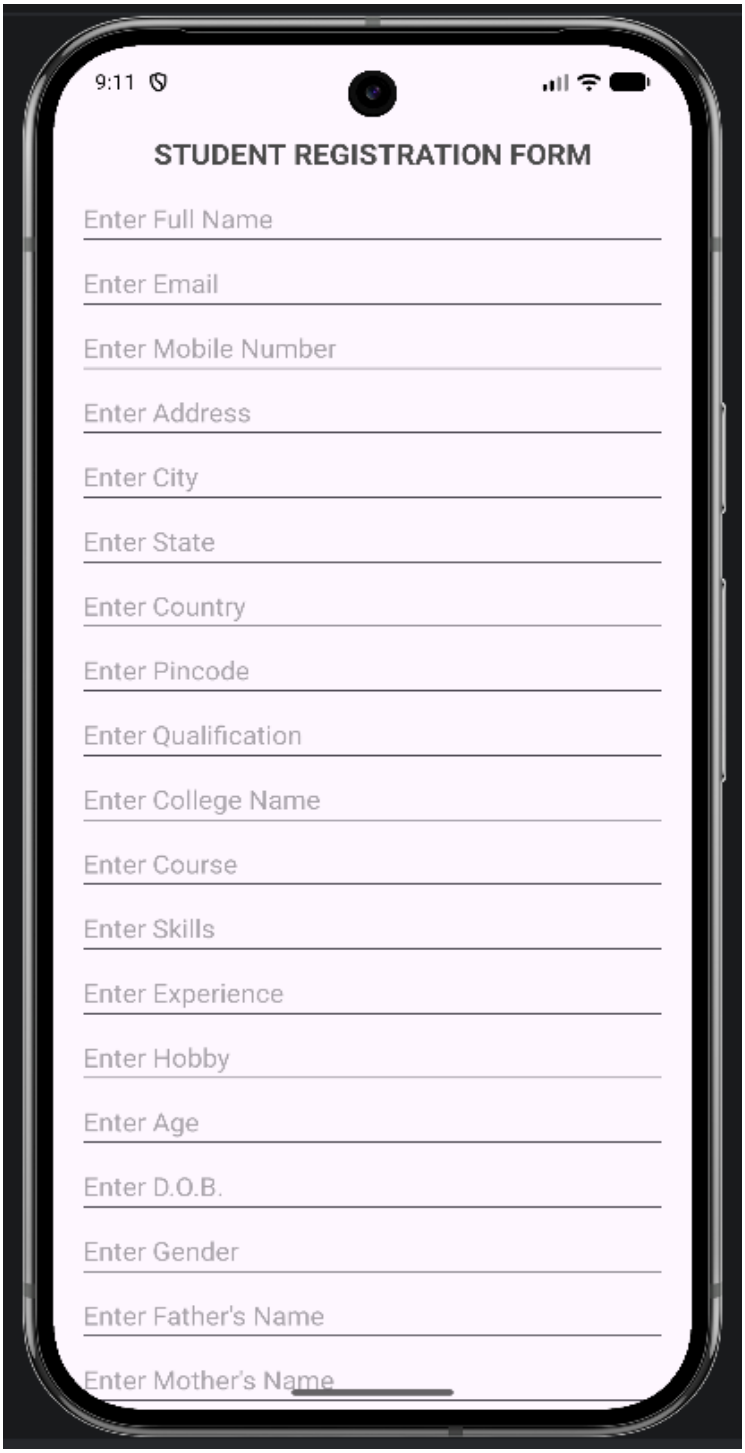
<!-- Submit -->
<Button
    android:id="@+id/button"
    android:layout_width="match_parent"
    android:layout_height="45dp"
    android:layout_marginTop="12dp"
    android:text="Submit"
    android:textAllCaps="false"
    android:textColor="@android:color/white"
    android:backgroundTint="#6B4EAC" />

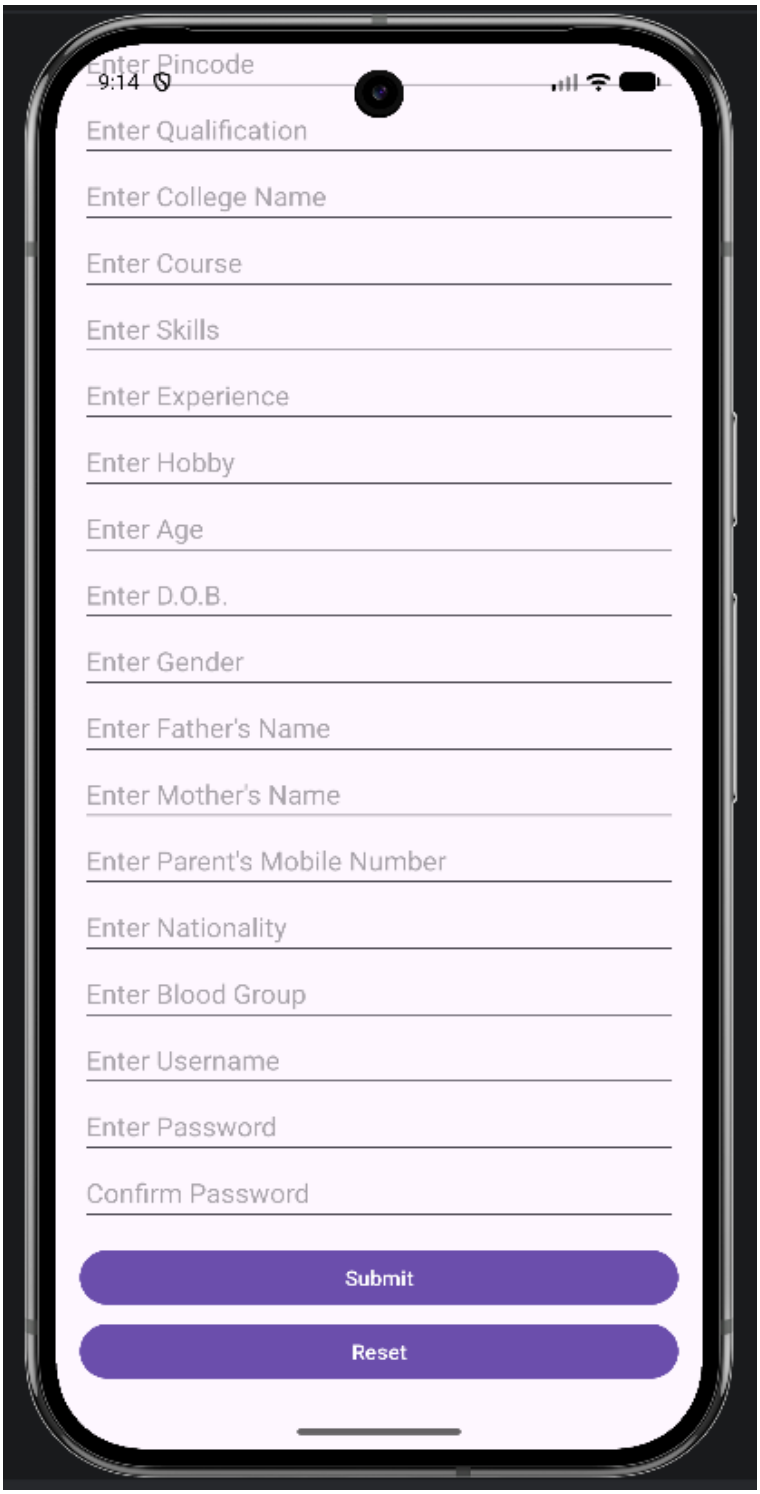
<Button
    android:id="@+id/button3"
    android:layout_width="match_parent"
    android:layout_height="45dp"
    android:layout_marginTop="5dp"
    android:text="Reset"
```

```
        android:textAllCaps="false"
        android:textColor="@android:color/white"
        android:backgroundTint="#6B4EAC" />

  </LinearLayout>
</ScrollView>
```

Output: ScrollViewDemo




9:14
Enter Pincode
Enter Qualification
Enter College Name
Enter Course
Enter Skills
Enter Experience
Enter Hobby
Enter Age
Enter D.O.B.
Enter Gender
Enter Father's Name
Enter Mother's Name
Enter Parent's Mobile Number
Enter Nationality
Enter Blood Group
Enter Username
Enter Password
Confirm Password
Submit
Reset

B) Develop an Android application using AbsoluteLayout demonstrating fixed positioning.

Project Name: AbsoluteLayoutDemo

Program Code:

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

   <TextView
      android:layout_width="232dp"
      android:layout_height="wrap_content"
      android:layout_x="60dp"
      android:layout_y="30dp"
      android:text="AbsoluteLayout Demo"
      android:textSize="20sp"
      android:textStyle="bold"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent" />

   <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_x="30dp"
      android:layout_y="100dp"
      android:text="Name:"
      android:textSize="20sp" />

   <EditText
      android:id="@+id/editTextText"
      android:layout_width="198dp"
      android:layout_height="wrap_content"
      android:layout_x="100dp"
      android:layout_y="90dp"
```

```
    android:ems="10"
    android:hint="Enter Name"
    android:inputType="text" />

  <TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_x="30dp"
    android:layout_y="170dp"
    android:text="Email:"
    android:textSize="16sp" />

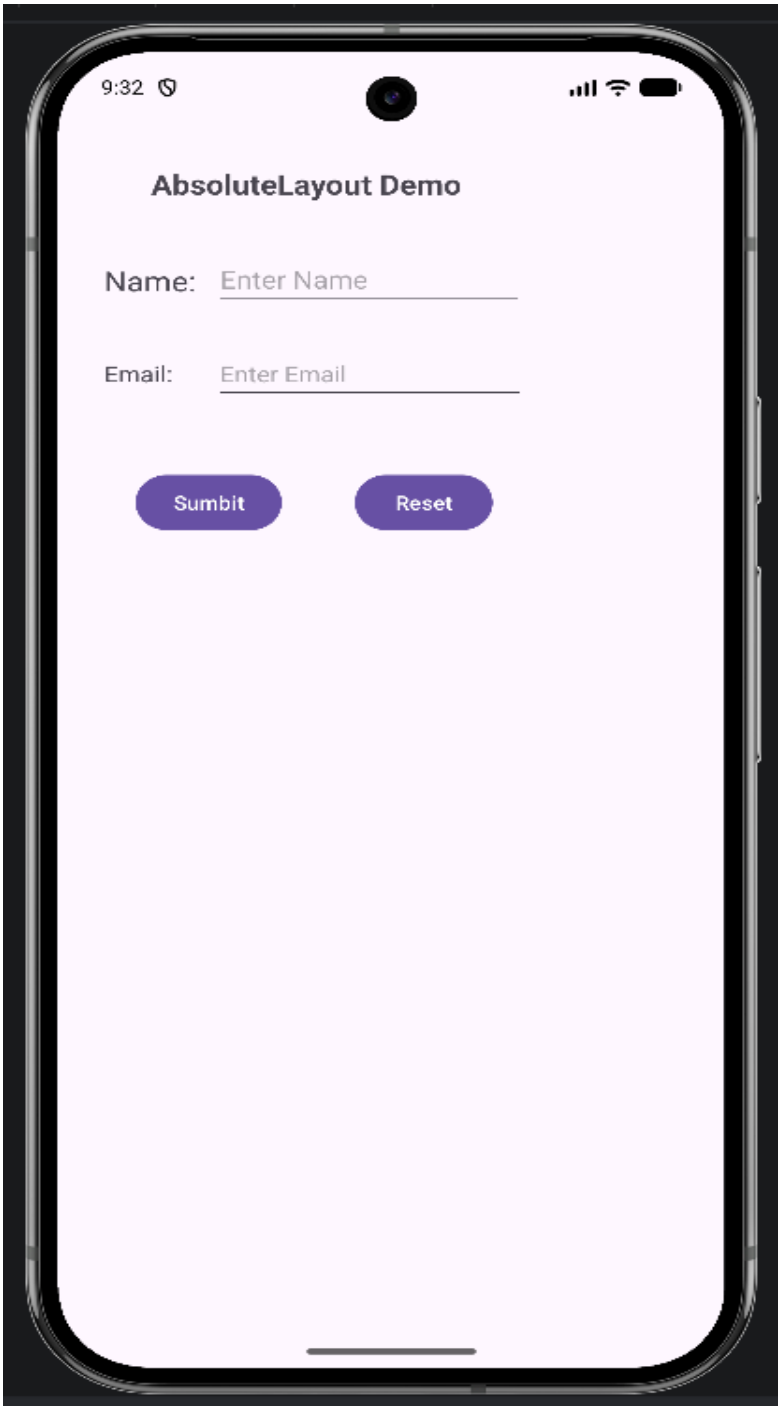
  <EditText
    android:id="@+id/editTextText2"
    android:layout_width="199dp"
    android:layout_height="wrap_content"
    android:layout_x="100dp"
    android:layout_y="160dp"
    android:ems="10"
    android:hint="Enter Email"
    android:inputType="text"
    android:textSize="16sp" />

  <Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_x="50dp"
    android:layout_y="250dp"
    android:text="Sumbit" />

  <Button
    android:id="@+id/button2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_x="190dp"
    android:layout_y="250dp"
    android:text="Reset" />

</AbsoluteLayout>
```

Output: AbsoluteLayoutDemo



C) Develop an Android application using FrameLayout to overlay UI elements.

Project Name: FrameLayoutDemo

Program Code:

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

   <ImageView
      android:id="@+id/backgroundImage"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:contentDescription="@string/background_description"
      android:scaleType="centerCrop"
      android:src="@drawable/onepiece" />

   <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_gravity="top|start"
      android:layout_marginStart="80dp"
      android:layout_marginTop="200dp"
      android:gravity="center_horizontal"
      android:text="Welcome To Frame Layout"
      android:textColor="#F22626"
      android:textSize="24sp"
      android:textStyle="bold" />

   <Button
      android:id="@+id/button"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
```

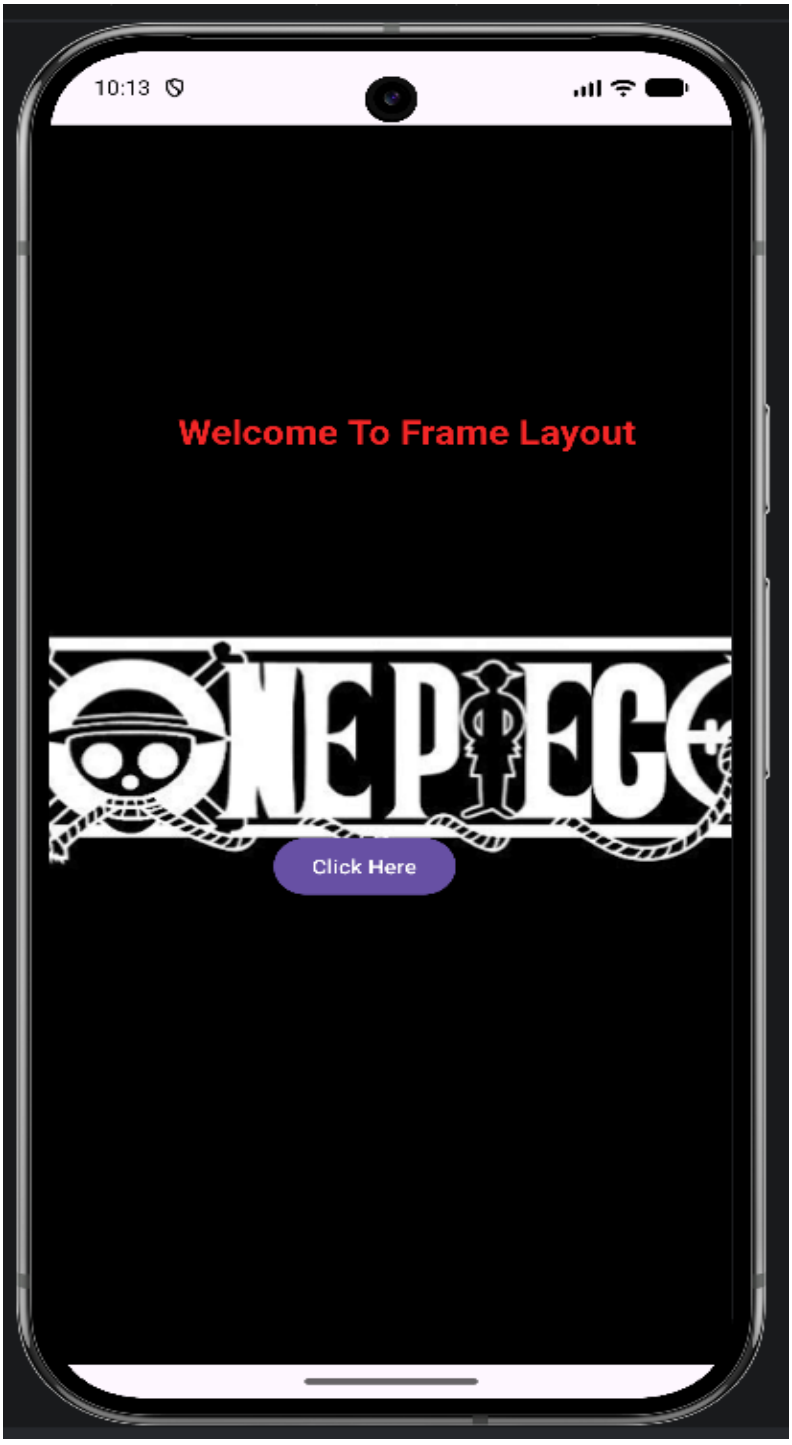
```
    android:layout_gravity="top|start"
    android:layout_marginStart="140dp"
    android:layout_marginTop="500dp"
    android:text="Click Here" />

</FrameLayout>
```

Drawable image file:
Filename: Onepiece.img



Output: FrameLayoutDemo




9:52
Nothing Happened
Click Here